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Research paper

Modelling the dependence structure of Y-STR haplotypes using graphical models

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ABSTRACT

Many methods have been suggested for evaluating the evidential value of a matching Y-chromosomal DNA profile obtained from a biological stain associated with a crime scene and the Y-chromosomal DNA profile of a suspect. Most of these methods are based on estimating the population frequency of the Y-profile. The common independence assumption between loci for autosomal DNA profiles cannot be used for Y-chromosomal DNA profiles. In this paper we reconsider the problem of population frequency estimation by application of Bayesian networks and the Chow-Liu algorithm to model dependencies between loci. We found that the method based on the Chow-Liu algorithm performs almost as well as the discrete Laplace method. We have also made comparisons to the independence model and we have demonstrated once again that the independence method for Y-profiles cannot be supported.

1. Introduction

The common independence assumption between loci for autosomal DNA profiles cannot be used for Y-chromosomal DNA profiles [1]. Many methods have been suggested for evaluating the evidential value of a matching Y-chromosomal DNA profile obtained from a biological stain associated with a crime scene and the Y-chromosomal DNA profile of a suspect. These include the kappa (κ) method [2], the coalescent method [3], discrete Laplace method [4], the generalized Good method [5], and distribution of number of matching males [6].

The results from Andersen and Balding (2017) ("How convincing is a matching Y-chromosome profile?" on the distribution of number of matching males) [6] suggest that for new modern kits, population frequencies may not be the most informative statistic. This is because the population frequency is believed to vary greatly depending on the reference population with matches most likely to occur to an individual only in the tens of meioses away from the suspect.

Nevertheless we reconsider the problem of population frequency estimation in this paper as it contributes to both the understanding of previous methods (such as the discrete Laplace method) and also to the application of Bayesian networks and the Chow-Liu algorithm in forensic science. It may still have some practical value for some practitioners, or with profiles from older kits.

2. Method

2.1. Structure of the joint probability mass function

Constructing a statistical model for Y-STR haplotypes is equivalent to describing the joint probability mass function (jpmf)

$$\Pr(X_1, X_2, \ldots, X_L),$$

where X_i is a random variable representing the allele(s)[1] at the ith locus. Frequencies can then be obtained by plugging in alleles $x_1, x_2, \ldots, x_L$ into this function

$$f(x_1, x_2, \ldots, x_L) = \Pr(X_1 = x_1, X_2 = x_2, \ldots, X_L = x_L),$$

and obtaining the numerical value.

There are many ways to model the jpmf. In this paper, we focus on the structural form of the jpmf. One such way is to assume

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[1] At most loci a haplotype exhibits only one allele per locus. Occasionally a locus may exhibit 0, 2, or 3 alleles such as for example the duplications at DYS385a/b.

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independence between loci to obtain

$$\Pr(X_1 = x_1, X_2 = x_2, \ldots, X_L = x_L) \overset{\text{indep.}}{\approx} \prod_{i=1}^{L} P(X_i = x_i).$$

For Y-STR profiles the assumptions used to obtain this approximation do not hold [1] and this approach cannot be supported. Without assuming independence the (general) multiplication law of probability allows us to express the joint probability function as a product of conditional probabilities. That is, for any events A, B and C the joint probability $\Pr(A, B, C)$ can be written

$$\begin{aligned}\Pr(A, B, C) &= \Pr(A)\Pr(B, C|A)\\ &= \Pr(A)\Pr(B|A)\Pr(C|A, B)\end{aligned}$$

This expression can be extended for any number of events. Applying this to the problem at hand we have

$$\begin{aligned}\Pr(X_1, X_2, \ldots, X_L) &= \Pr(X_1)\Pr(X_2, \ldots, X_L|X_1)\\ &= \Pr(X_1)\Pr(X_2|X_1)\Pr(X_3, \ldots, X_L|X_1, X_2)\\ &= \Pr(X_1)\Pr(X_2|X_1)\Pr(X_3|X_1, X_2)\Pr(X_4, \ldots, X_L|X_1, X_2, X_3).\end{aligned}$$

Factors, such as $\Pr(X_4,\ldots,X_L|\ X_1, X_2, X_3)$, in this expansion which condition on many variables are undesirable because they are difficult to estimate from any reasonable number of empirical data. The difficulty in estimation arises because, as the number of dimensions in the contingency table increases, there are insufficient data to adequately populate the cells of the table. This in turn means that there are many zero entries and that the estimates are highly variable. There is a general trade-off in modeling: more flexible models (in this case high dimensional contingency tables or factors with many conditional variables) describe the observed data better, but are generally poorer when it comes to prediction. This well-known phenomenon is sometimes called over-fitting.

The independence assumption implies that the conditioning does not matter. That is, if X_1 and X_2 are independent, then

$$\Pr(X_2|X_1) = \Pr(X_2)$$

(and so on). If every pair of loci is assumed to be independent then the *product-rule* estimator is obtained. This is over-simplification for Y-STR. The model is now very simple in that only one dimensional tables are used, but the model is now too simple in that it fails to include potentially large dependencies.

We find ourselves stuck between two extremes; assuming full independence results in a model that is too simplistic, and assuming full dependence results in a situation where we do not have sufficient data to fit the model. We opt here for a compromise strategy – namely to find a model somewhere in between these extremes; the model should not be too simple (independence assumption), nor be too complicated (full jpmf where there is no conditional independence between any pair of random variables).

Take the following example with $L = 4$ loci. Let us assume that

$$\Pr(X_3|X_1, X_2) = \Pr(X_3|X_2).$$

That is, we assume that X_3 is independent of X_1 given X_2. Furthermore if we assume that

$$\Pr(X_4|X_1, X_2, X_3) = \Pr(X_4|X_2),$$

i.e. that X_4 is independent of X_1 and X_3 given X_2, then

$$\Pr(X_1, X_2, X_3, X_4) = \Pr(X_1)\Pr(X_2|X_1)\Pr(X_3|X_1, X_2)\Pr(X_4|X_1, X_2, X_3) \quad (1)$$

$$\overset{\text{by assump.}}{=} \Pr(X_1)\Pr(X_2|X_1)\Pr(X_3|X_2)\Pr(X_4|X_2). \quad (2)$$

This is one such example that is somewhere between the full jpmf and the independence assumption. Ideas similar to this have previously been considered briefly without much success [7,8].

The dependence structure of any joint probability distribution may be represented as a graph. Each random variable in the distribution appears as a node in the graph. Dependencies between pairs of random variables are represented as edges between the nodes representing those variables. If the edges in the graph are undirected, then the graph describes a Markov Random Field, and there is considerable literature devoted to this [9]. However, if the edges are directed, and there are no cycles, then we find ourselves in the field of Bayesian networks [10].

Bayesian networks (BNs) (also known as directed acyclic graphs (DAGs), or graphical models) can be used for both illustrating and reasoning with conditional dependencies, and have a strong presence in modern forensic literature [11]. Again, each random variable is drawn as a node in the graph. There is a directed edge from X_i to X_j if X_j is dependent on X_i (i.e. $P(X_j|X_i) \neq P(X_j)$), and X_i is said to be a parent of X_j. The term *acyclic* refers to the fact that the graph has no *cycles*. That is, there must be no *path* – sequence of edges – that allows the dependence to return to X_i. The directedness of the graph, and the acyclic restriction means that the jpmf can be represented by the equation

$$P(X_1, X_2, \ldots, X_L) = \prod_{i=1}^{L} P(X_i|\text{pa}(X_i)),$$

where $\text{pa}(X_i)$ denotes the parents of X_i. This representation says that given its parents, the random variable X_i is conditionally independent of all other variables in the distribution. This is a powerful statement which often allows complex problems to become computationally feasible. Note, that it is not allowed to have directed edges in both directions (both from X_i to X_j and from X_j to X_i) as this introduce a cycle of length two.

Using this notation, the independence model consists of the graph with nodes $X_1, X_2, \ldots, X_L$ and no edges. The full jpmf (with no conditional independence between any pair of nodes) for four variables can be represented by the DAG shown in Fig. 1. The jpmf from (2) can then be represented by the DAG shown in Fig. 2.

Lastly, note that some factorisations and thereby DAGs are equivalent in the sense that they look different but actually represent the same distribution (the same probability mass function). For example, one factorisation of $\Pr(X_1, X_2, X_3, X_4)$ based on Fig. 1 is $\Pr(X_1)\Pr(X_2|X_1)\Pr(X_3|X_1, X_2)\Pr(X_4|X_1, X_2, X_3)$. An equivalent factorisation is $\Pr(X_4)\Pr(X_3|X_4)\Pr(X_2|X_3, X_4)\Pr(X_1|X_2, X_3, X_4)$ (DAG not shown).

2.2. The Chow-Liu method

The Chow-Liu (CL) algorithm [12] provides a structured way to approximate the jpmf by only using first-order dependencies (single columns of two-dimensional tables, so for example $\Pr(X_2|X_1)$, $\Pr(X_3|X_2)$). The Chow-Liu algorithm builds a *tree* using the mutual information for pairs of random variables. The mutual information is a formal measure from information theory which quantifies the degree of

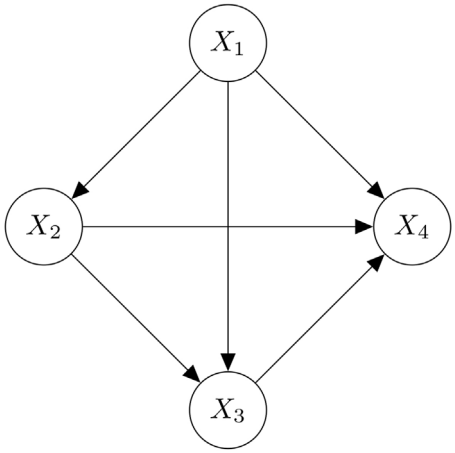


Fig. 1. Representing the full jpmf $\Pr(X_1, X_2, X_3, X_4) = \Pr(X_1)\Pr(X_2|X_1)\Pr(X_3|X_1, X_2)\Pr(X_4|X_1, X_2, X_3)$ (with no conditional independence) as a DAG.

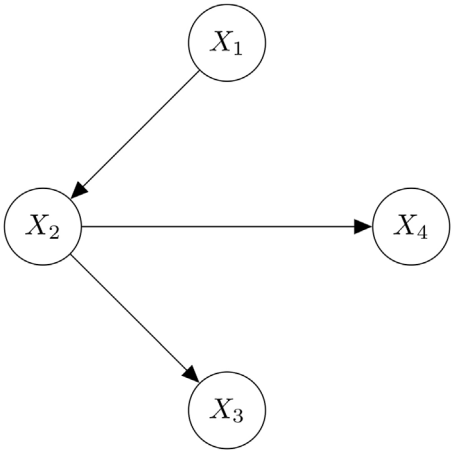


Fig. 2. Representing $\Pr(X_1, X_2, X_3, X_4) = \Pr(X_1)\Pr(X_2|X_1)\Pr(X_3|X_2)\Pr(X_4|X_2)$ from (2) as a DAG.

dependence between two random variables. For two discrete random variables X (taking values x_i for $i = 1, 2, \ldots, n_X$) and Y (taking values y_j for $j = 1, 2, \ldots, n_Y$), the mutual information between X and Y is given by

$$I(X, Y) = \sum_{i=1}^{n_X}\sum_{j=1}^{n_Y} \Pr(X = x_i, Y = y_j)\log\left(\frac{\Pr(X = x_i, Y = y_j)}{\Pr(X = x_i)\Pr(Y = y_j)}\right).$$

A *tree* is a special type of graph which contains exactly one path between every pair of nodes. A *connected*[2] acyclic graph is therefore a tree. The Chow-Liu algorithm works by sequentially adding edges between pairs of variables based on a descending set of pairwise mutual information values. Edges are added until a tree is formed. It can be shown ([12]) that the resulting tree is the tree amongst all possible trees with the smallest Kullback-Leibler (KL) divergence [13]. This means that the resulting tree is the best approximation (at least according to the KL metric) that you can hope to get using only first-order dependencies.

Several publications [14,15] have previously investigated the use of mutual information for Y-STRs. The idea is that mutual information is one way to measure the degree of dependency between (here, two) variables. The mutual information is always non-negative (that is zero or positive). If the mutual information is high, then it means that knowing the value of one variable gives a lot of information about the other; that is, they are highly dependent. The traditional Pearson (product-moment) correlation is another, complementary measure of (linear) dependency between variables. The correlation coefficient is not directly related to the mutual information, but the concept is similar in that it assigns a numerical measure that quantifies the dependency between two variables.

Note, that the Chow-Liu method does not give an estimated probability that is equal to the sample frequency. That is, if a sample (database) consists of n haplotypes, and we are considering a singleton,[3] then the sample frequency is $1/n$. The estimate arising from a tree constructed using the Chow-Liu algorithm however is not necessarily $1/n$. In fact it is quite likely that the estimate will be something other than $1/n$. However, if the full empirical jpmf was used, the estimated probability would be the same as the database frequencies. Therefore one way to think of the full empirical jpmf is that it contains all information of the sample which was used to estimate the tables, and it only contains this information. In our application, we want to know about population frequencies – not sample frequencies – so we need a model that generalizes better. The Chow-Liu algorithm is one way to obtain such models. It should also be noted that the Chow-Liu algorithm does not exploit our knowledge of population genetics. However, if the measure of success is the accuracy (and precision) of the approximation, then practical utility might outweigh theoretical concerns.

2.2.1. Example

We demonstrate the tree-building phase of the CL algorithm using the Danish reference database [16]. We only require estimates of the mutual information to be able to construct the tree. The database has been restricted to PowerPlex Y profiles with integer alleles and treating DYS385a/b as two loci (cf. below in Section 2.3).

There are 12 ($L = 12$) loci. Hence, there are $L(L-1)/2 = 66$ pairs of loci ignoring order. The algorithm proceeds by calculating the mutual information [12,14] for each of these 66 pairs. The mutual information for all of the locus pairs needed to produce the final tree are listed in Table 1.

The method is now to include edges between the nodes (loci) in the list, starting from the top of the list and moving downwards. To begin with, the edges have no direction. The exception is that an edge is not added if it introduces a cycle (a cycle is a path from a node to itself using each edge on the path only once). Note that a cycle for an undirected graph is different from a cycle in a directed graph, in that the direction of the edges restricts what constitutes a cycle. When all possible edges are added, the edges can be directed. The directions of the edges do not matter because the graph is a tree, and therefore they can be chosen randomly. So for this example, the first edges added are DYS392–DYS438, and then DYS438–DYS437, and then DYS385a–DYS438. The next possible edge is DYS385a–DYS437 but it would introduce a cycle (DYS385a–DYS437–DYS438–DYS385a) and hence it is skipped.

This procedure continues until $L - 1$ edges have been added. (It can be shown that a tree with L nodes has precisely $L - 1$ edges.) Adding an Lth edge would always result in a cycle.

For this example, the final graph is shown in Fig. 3.

2.3. Simulation study

It now falls upon us to demonstrate the performance of the resulting approximation. To that end, we performed a simulation study using R [17] and the R package `malan` [18].

We simulated 20 different genealogies over 250 generations. The genealogies differ by varying key parameters. We considered two growth rates; one where the growth rate was zero, and hence the initial population size is equal to the final population size, and one where there was 2% growth per generation. We set up the simulation in the latter so that the initial population size lead to a final population size that was approximately the same as the zero growth simulation.

2.3.1. Variances in reproductive success

We followed Andersen and Balding [6], in that we also considered different variances in reproductive success (VRS). VRS is a measure of variance in relative paternity probabilities to allow some males to have a higher probability of becoming a father than others. It is easier to model the relative paternity probabilities, but they can be converted to paternity probabilities by normalisation. And relative paternity probabilities can be obtained from paternity probabilities by dividing by the average of all paternity probabilities. Hence, the mean of the relative paternity probabilities is one.

We modelled the relative paternity probabilities, X, as independent Gamma random variables with mean 1. This is parameterised as $X \sim \text{Gamma}(\alpha, 1/\alpha)$ (shape $= \alpha$, scale $= 1/\alpha$) to obtain mean one. When normalising these independent Gamma variables, a symmetric Dirichlet distribution with parameter α is obtained.

To explain the VRS term, consider Y, the number of sons each father has. A man's total number of sons is, to a close approximation when N is large, Poisson distributed. If $X \sim \text{Gamma}(\alpha, \beta)$ (shape $= \alpha$, scale $= \beta$,

[2] Connected means there is a path between every set of nodes, and that there are no unreachable nodes.

[3] A singleton is a haplotype that only occurs once in the database.

Table 1
The locus pairs and the mutual information from the Danish database example needed to build the tree. The "DYS" prefix on locus names has been omitted for brevity. The "MI" column refers to the mutual information between "Locus1" and "Locus2". The column "Cycle?" has an "X" if adding this edge would introduce a cycle (such edges are not added).

Locus1	Locus2	MI	Cycle?	Locus1	Locus2	MI	Cycle?
392	438	0.618		389I	390	0.202	X
438	437	0.553		439	437	0.201	X
385a	438	0.513		390	392	0.195	X
385a	437	0.408	X	19	389II	0.190	X
390	438	0.401		392	439	0.188	X
392	437	0.398	X	391	392	0.179	X
392	385a	0.394	X	391	437	0.179	X
390	437	0.369	X	385a	439	0.178	X
390	385a	0.367	X	19	439	0.175	X
389I	437	0.325		389I	392	0.174	X
389I	385a	0.314	X	19	385b	0.174	X
439	438	0.257		389II	438	0.172	X
391	438	0.257		385b	438	0.169	X
389I	438	0.256	X	19	385a	0.168	X
391	385a	0.254	X	389II	390	0.164	X
19	437	0.239		19	392	0.162	X
385a	385b	0.232		390	391	0.159	X
19	438	0.207	X	389II	385a	0.158	X
389II	437	0.202		393	438	0.149	

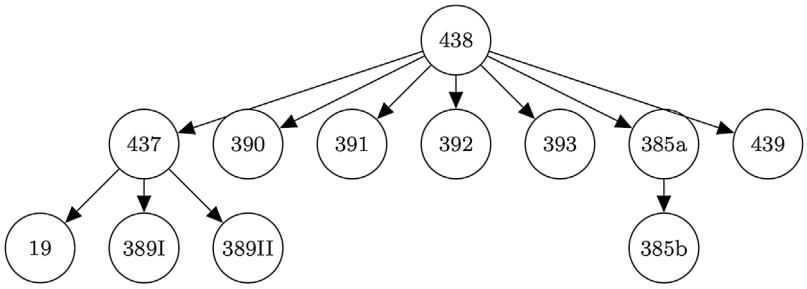


Fig. 3. The resulting DAG from using the Chow-Liu method on the Danish reference database. "DYS" prefix on locus names removed.

mean $= \alpha/\beta$) and $Y|X = x \sim \text{Poi}(x)$, then Y has a Gamma-Poisson/negative binomial distribution with variance $\beta^2(\alpha(1 + 1/\beta))$. Recall that we used $\beta = 1/\alpha$ to get a mean relative paternity probability of one. This results in a total variance of the number of sons is $1 + 1/\alpha$. The value of VRS is this total variance minus one, or in other words VRS $= 1/\alpha$. Hence, the greater the value of α, the lower the total variance of the number of sons is.

For example, consider a value of $\alpha = 5$. Then, the total variance of the number of sons is 1.2. Hence for $\alpha = 5$, VRS $= 1.2 - 1 = 1/5 = 0.2$. Futher, when $\alpha = 5$, a man's relative paternity probability has 95% probability to lie between 0.32 and 2.05 (quantiles of the Gamma distribution with shape parameter $\alpha = 5$ and scale parameter $1/\alpha = 0.2$). Thus, some males are twice as likely to father son(s) than the average father (with relative paternity probability of one).

The standard Wright-Fisher model assigns all N men the same relative probability of paternity (equal to one), and so there is no variance and we label this VRS $= 0$. This corresponds to the limiting case with "$\alpha = \infty$".

Here, we considered both values VRS $= 0$ ("$\alpha = \infty$") and VRS $= 0.2$ ($\alpha = 5$).

2.3.2. Population size

The simulations were set up so that the second-to-last generation was of size 500,000. The last three generations constituted the live population. The initial population size in the situation where we allowed 2% growth was 3682 and this gave a live population of size 1,500,196. The live population size was 1.5 million under the zero growth model. Thus, the minimum population frequency was $1/(1.5 \times 10^6) \approx 6.67 \times 10^{-7}$.

Two growth rates and two VRS values gives four parameter sets. We simulated five genealogies for each set of parameters. In total this gave 20 different genealogies.

2.3.3. Kits

We considered two Y-STR profiling kits: Minimal (9 loci) and PowerPlex Y (12 loci). We used the mutation count data from YHRD.org release 55 [19] with a Beta(1.5, 200) prior distribution similar to [6]. For each simulated genealogy, we sampled mutation rates from the posterior distribution obtained from the Beta prior updated with binomial mutation count. This was done five times.

Both kits contain copies of the duplicated locus DYS385. In our simulation study, DYS385a and DYS385b are treated as two independent loci, whereas in practice the order of the duplicated locus is unknown. This means, for example that the allele pair 13/14 cannot be distinguished from the allele pair 14/13. We use the same mutation rate for DYS385a and DYS385b, estimated by halving the reported mutation count for locus DYS385.

2.3.4. Founder haplotypes and allelic ladder

This simulation process is very similar to that of [6]. Two changes have been made to the R package `malan` [18] post the publication of [6]. These are (Andersen and Balding, preprint, https://doi.org/10.1101/373423):

1. A Danish reference database [16] was used to randomly choose founder haplotypes (according to their frequency in the database) instead of all founders having the same haplotype.
2. Also, an allelic ladder with reflecting boundaries was used (the ladder size was chosen to reflect the number of integer alleles observed in YHRD.org release 55 [19]). Therefore, the allele can only mutate upwards (downwards) when it is at the lower (upper) boundary.

2.3.5. Cases

Fifty (50) cases were sampled with two different database sizes (500 and 1000) for each live population with haplotypes. A case consisted of a database and then an additional individual (the suspect). The population frequency of the suspect's haplotype was saved. In addition, we estimated the population frequency of the suspect's profile using both the Chow-Liu method [12] and the discrete Laplace method [4] based on the extended database (database with suspect's profile added). There are two steps in the Chow-Liu method. The first of these is the determining the Chow-Liu tree (the factorisation) and this requires only estimates of the one- and two-way probability tables to estimate the mutual information. The second step uses the factorisation implied by the tree to estimate the joint probability by a number of conditional probabilities that need to be estimated. We use maximum likelihood estimation for the second step. This is implemented in the `bn.fit` function in the `bnlearn` [20] package. The discrete Laplace method requires that a number of clusters be specified. We used a maximum of 10 and 15 clusters of databases of size 500 and 1000, respectively.

Recall that the discrete Laplace method works by trying to estimate parameters of a finite mixture model (consisting of discrete Laplace components) that best fits the observed haplotypes in the reference database. This requires that we define our concept of *best* fit. In this study, the fit with the lowest value of the Bayesian Information Criterion (BIC) [21] was used to estimate the population frequency of the suspect's profile. We denote this value "*min BIC*". In addition, we also estimated the suspect's profile probability for each number of clusters and chose the one with the highest probability. This was meant to serve as a conservative estimate. We denote this value '*max prob.*".

Twenty thousand (20,000) cases were simulated in total: 2 growth rates × 2 VRS values × 5 genealogy repetitions × 2 kits × 5 mutation rates draws × 2 database sizes × 50 cases).

A combination of the R packages `bnlearn` [20] and `gRain` [22] were used to estimate the frequencies using the Chow-Liu method. The R package `disclapmix` [23,24] was used to estimate the frequencies using the discrete Laplace method [4].

2.4. Error summary

We used the mean squared error (MSE) of the match probabilities on the $\log_{10}$ scale to summarise the error in an estimator. In other words, let $i = 1, 2, \ldots, n$ be the case number, p_i the true population frequency and $\hat{p}_i$ the estimated frequency by an estimator. We then define

$$\mathrm{MSE}(\hat{p}) = \frac{1}{n}\sum_{i=1}^{n}(\log_{10}(p_i) - \log_{10}(\hat{p}_i))^2 = \frac{2}{n}\sum_{i=1}^{n}\log_{10}\frac{p_i}{\hat{p}_i}.$$

3. Results

All analyses were performed in R [17].

Kernel density estimates of the distribution of the singleton proportions of the databases sampled (without the suspect included) are shown in Fig. 4. As expected, the singleton proportion increases with smaller databases and the number of loci. This is because the haploptype diversity increases with an increase in the number of loci. We also note that increasing the reproductive success decreases the proportion of singletons. This again is unsurprising as increased reproductive success for a male means more chances to pass his haplotype along.

The estimates are compared to the true, known population frequency in Fig. 5. A more detailed version of this figure is given in Fig. A1. The mean squared error between the frequency estimate on $\log_{10}$ scale and the population frequency on $\log_{10}$ scale, as described in Section 2.4, is given in Table 2 for each combination of database size and estimation method. Both the Chow-Liu (CL) based estimator and the discrete Laplace (DL) based estimator would place probability mass on haplotypes which may never be observed in reality. As a consequence we know, a priori, that these estimators will, in general, under-estimate the true population frequency for existing types. Equally, we know that the degree of under-estimation will increase as the number of loci increases, because of the increase in the number of potential haplotypes. Both of these facts are evident in Fig. 5. We also note that neither method dominates. That is, neither the CL or DL estimator is noticeably better or worse than the other.

This means that all cases can be compared to check how the two methods agree. This is depicted in Fig. 6.

Fig. 6 shows the identity line ($y = x$), the estimated regression line, and a 95% prediction interval. We discuss each of these in turn. The identity line shows where we expect the results to lie (on average), if the DL and CL produce very similar estimates. The regression lines (which have been forced through zero) show the estimated relationship between the DL and CL estimates. If the slope is greater than one, then this is evidence that the DL produces slightly higher estimates (i.e. under-estimates slightly less) than the CL method. This appears to be the case, however we should not read too much into this, as the prediction interval (dotted line) includes the identity line. This means that it the DL might produce a higher value or it might produce a lower value.

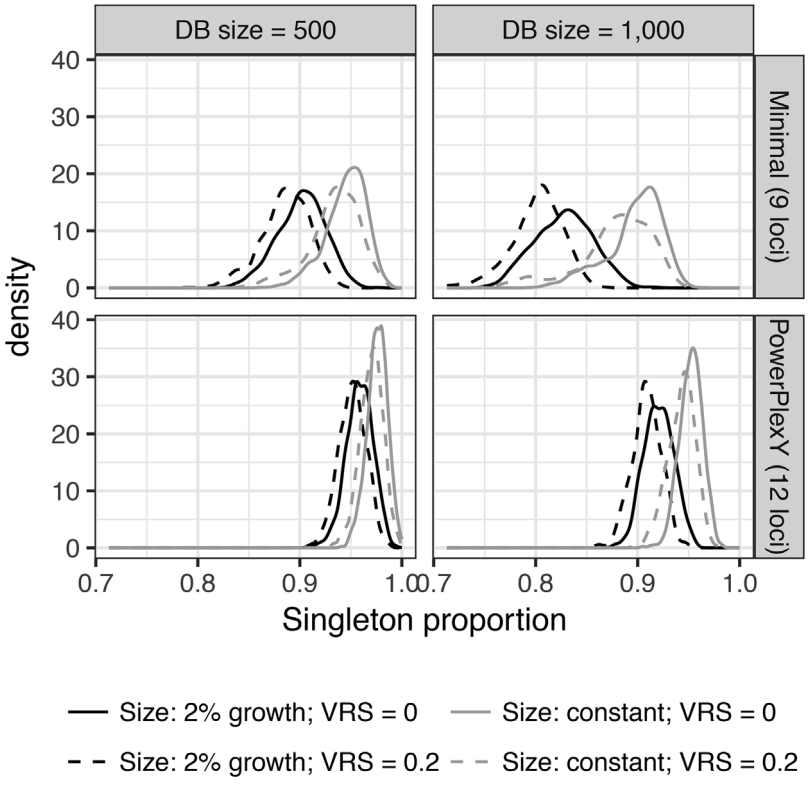


Fig. 4. Distribution of singleton proportions of the databases sampled (without the suspect included).

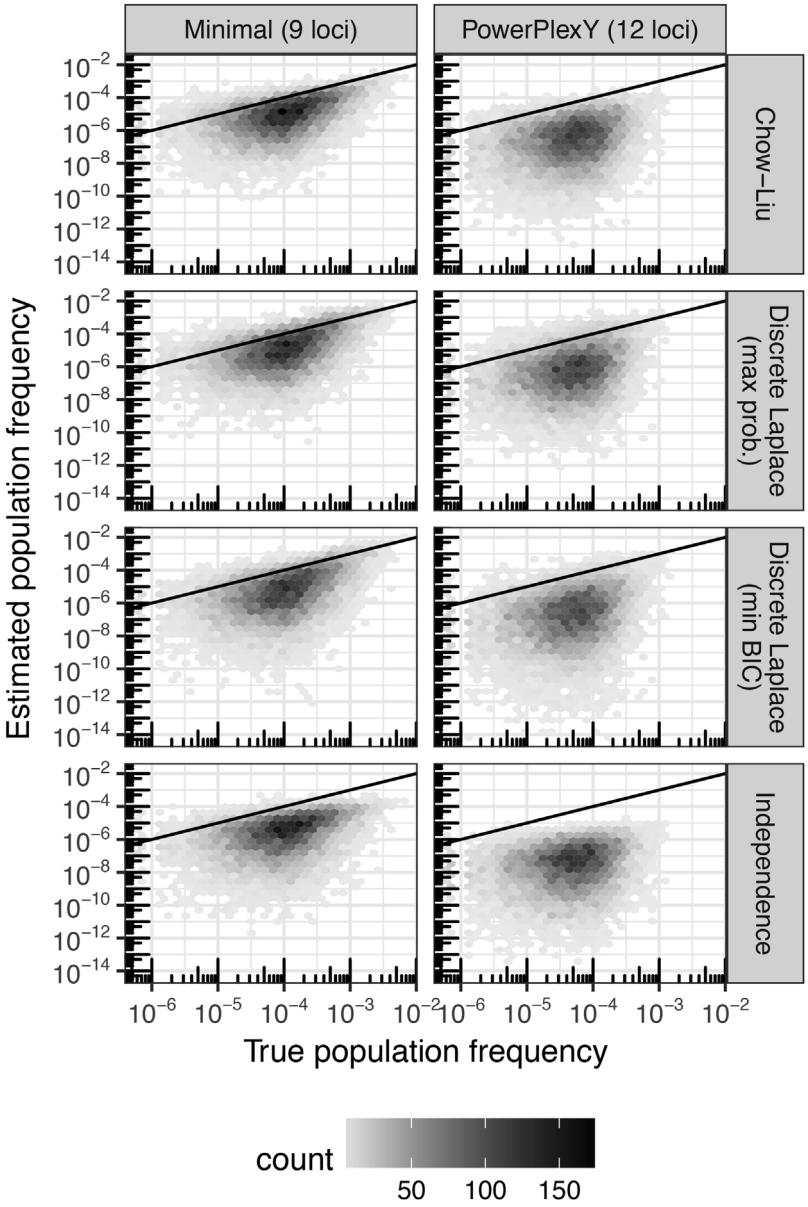


Fig. 5. Comparison of the estimators to the known population frequencies. The superimposed identity line ($y = x$) is shown. A more detailed figure is available in Fig. A1.

Table 2
The mean squared error between the frequency estimate on $\log_{10}$ scale and the population frequency on $\log_{10}$ scale as described in Section 2.4.

DB size	Estimator	Minimal (9 loci)	PowerPlexY (12 loci)
500	Chow-Liu	2.22	6.76
500	Discrete Laplace (max prob.)	2.25	5.64
500	Discrete Laplace (min BIC)	3.55	9.03
500	Independence	4.22	12.66
1,000	Chow-Liu	2.51	7.40
1,000	Discrete Laplace (max prob.)	2.24	5.79
1,000	Discrete Laplace (min BIC)	3.31	8.22
1,000	Independence	4.29	12.71

4. Discussion

In Fig. 5, the estimates are compared to the true, known population frequency. Note, that both estimators in general give a lower estimate than the true population frequency. This is not that surprising. A maximum of 1,500,000 haplotypes exist in the live population we used (and in reality a lot fewer as some individuals share the same haplotype). However, both models must assign probabilities to a much, much larger number of haplotypes. The methods do not know which haplotypes exist and which do not exist, and when the methods assign probability mass to haplotypes that actually do not exist, they are both required – in general – to under-estimate the frequency for existing types as only a total probability mass of 1 can be used.

Table 2 shows the MSE values, as described in Section 2.4, calculated for each combination of the database size and the estimator. There are several points that need to be highlighted. First, the error increases with respect to number of loci. Similarly, all four MSE values increase with database size for the Chow-Liu and Independence methods. The MSE values for Discrete Laplace methods, however, *decrease* with an increase in database size – the exception being the max prob. for PowerPlexY.

Having a statistical model of the haplotypes, that is one which assigns a probability to unobserved haplotypes, makes it possible for one to carry out mixture analysis as demonstrated in [25,26].

Fig. 6 shows that CL and DL methods have a high degree of

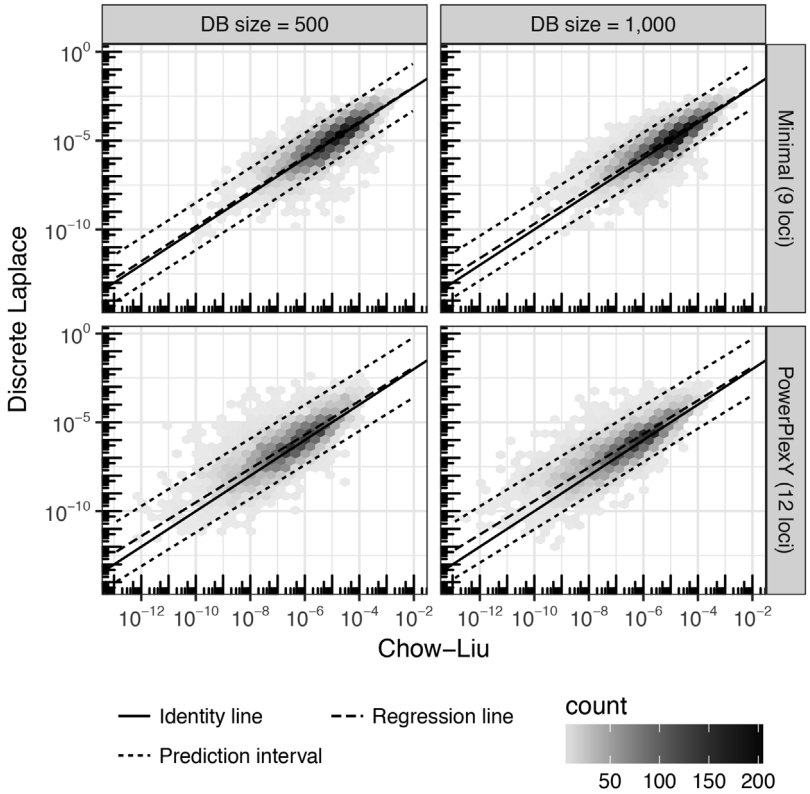


Fig. 6. Comparison of the estimates by the Chow-Liu method and by the discrete Laplace method. The superimposed identity line ($y = x$), the estimated regression line, and a 95% prediction interval are shown.

agreement, and that it is difficult to say one out-performs the other in any way.

Developing testing regimes in many modern DNA situations is problematic. The issue arises from the inability to know the true probability of a genotype, in the case of autosomal profiles, or a haplotype, in the case of lineage (e.g. Y, mitochrondrial) profiles. This is because all profiles are sufficiently discriminatory, and hence sufficiently rare, that we will never be able to collect enough data to adequately inform the probabilities. Discrimination is one issue, the fact that the population is continually changing (through births and deaths) is another. We therefore, in this paper, restrict our performance measures to the relativistic situation – that is, we compare two estimators under identical simulated situations. This may limit the conclusions we draw, but in the absence of sensible alternatives, we still think that the results provide information for practitioners and researchers.

In Section 2.2.1 we applied the Chow-Liu method to a Danish reference database [16]. Two questions naturally arise from this: "What is the impact of considering DYS385a/b as two independent markers?" and "Does the structure differ between different populations?" These are two interesting questions, but also two questions that are difficult to answer with great certainty. To help answer these questions, we have used the PowerPlex Y part of the worldwide database published in [27]. We then applied the Chow-Liu method to each of the 5 continents (Africa, $n = 445$; Asia, $n = 3{,}458$; Europe, $n = 11, 968$; Latin America, $n = 1{,}183$; North America, $n = 2{,}576$).The results are shown in Fig. 7. It can be seen that all the graphs except the one for the European continent are the same. The difference between the two graphs are that in the European, DYS439, DYS392, DYS437, and DYS389I are not connected directly to DYS385 but instead to DYS385 via DYS438. Except from that, the structure is very similar. It can be seen by comparing Fig. 7 to Fig. 3 that DYS385a/b has a much more central role in Fig. 7. Again, we want to stress the fact that DYS385 is difficult to handle because it has duplications and that the allele pair 13/14 cannot be distinguished from the allele pair 14/13. This means that a "match" on DYS385 may not be a true match because both 13/14 and 14/13 are reported as 13/14.

4.1. Perspective

In our implementation of the Chow-Liu method we treated the variables (loci) as being categorical. We make this as a distinction over them being treated either as ordinal, or as quantitative variables. That is, the possibilities for each locus (the alleles) are simply treated as categories, and no account is taken of their value, or their occasional non-integer designation (e.g. the 11.2 allele at locus DYS391). It is possible that the estimation would benefit from calculating the mutual information using for example the discrete Laplace method. On the other hand, categorical distributions have the advantage that intermediate alleles can be included in the model.

There exists a generalisation to the Chow-Liu method such that instead of only a second order approximation (two dimensional contingency tables), a kth order approximation can be obtained. This generalisation is called t-cherry junction trees [28,29]. The second order t-cherry junction tree is actually the Chow-Liu tree. It would be interesting to see how it would perform. It is an inescapable truth that the higher the order of dependency that we consider, the more sparse the tables become. That is, we have insufficient data to populate all of the cells of the table, and as a result there are many cells which are zero. If we believe that there should be some probability mass assigned to every cell, and there are arguments for why and why not, then it may be prudent to consider smoothing the contingency tables or removing redundancies, and this in turn may improve the method. One way of addressing this in the Bayesian context is to assume a flat prior for all cells in the contingency tables.

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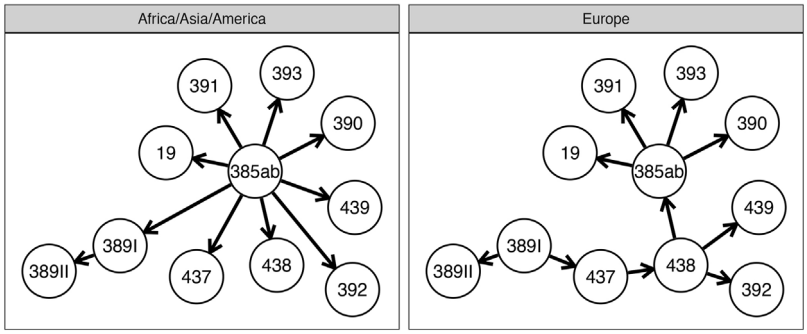


Fig. 7. Comparison of the structures resulting from using the Chow-Liu method on the continents' databases from [27].

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.fsigen.2018.07.014.

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